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Title: AI-Enhanced Blockchain Auditing: Improving Transparency and Security in Financial Systems

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Summary:

This paper investigates the integration of artificial intelligence (AI) with blockchain technology to enhance auditing processes in financial systems, focusing on improving transparency and security. The authors argue that combining AI's data processing capabilities with blockchain's immutable ledger creates a powerful framework for real-time audit trails, fraud detection, and compliance verification. The study emphasizes how AI can analyze blockchain-based financial transactions to identify anomalies and ensure data integrity, addressing the limitations of traditional auditing methods in handling decentralized and high-volume data environments.

The methodology involves a mixed-methods approach, including a simulation of AI-driven auditing on a blockchain platform using financial transaction data from 10 firms (2020-2023) and interviews with blockchain and audit experts. Results show that AI-enhanced blockchain auditing reduces fraud detection time by 35% and improves transaction verification accuracy by 20% compared to manual methods. A case study of a fintech company illustrates how AI identified unauthorized transactions on a blockchain ledger, preventing a potential \$1.5 million loss. Challenges include the complexity of integrating AI with blockchain systems, high computational costs, and ensuring regulatory compliance across jurisdictions.

The paper concludes that AI-blockchain integration offers a robust solution for transparent and secure financial auditing, particularly in industries like fintech and supply chain finance. However, it highlights the need for standardized protocols and auditor training to navigate these technologies effectively. The authors call for further research into scalable AI-blockchain frameworks and their long-term impact on audit reliability, positioning this approach as a critical step toward future-proofing financial systems against emerging risks.